Materials Science - Team Project

Subject: Material Modelling of Shape Memory Alloys (SMAs) and their Fatigue Behaviour

Motivation: Shape Memory Alloys (SMAs), particularly NiTi-based systems, have emerged as a class of smart materials with unique functional properties such as the shape memory effect and superelasticity. These phenomena arise from reversible martensitic phase transformations, enabling SMAs to recover large strains upon thermal or mechanical activation. Their exceptional combination of high recoverable strain, corrosion resistance, and energy absorption capacity has led to widespread applications in biomedical devices (stents, orthodontics), aerospace (morphing structures, actuators), and automotive systems (dampers, couplings).
However, the long-term reliability of SMA components is strongly influenced by functional fatigue under cyclic thermomechanical loading. Repeated actuation leads to permanent strain accumulation, reduction of recovery strain, and hysteresis shifts, which degrade performance and limit service life. Understanding and predicting this fatigue behaviour is therefore essential for the safe and efficient design of SMA-based systems.

Task Description: The aim of this task is to acquire knowledge and evaluate material models that capture both the constitutive behaviour of SMAs and their fatigue degradation mechanisms. The work will involve:

- Literature review of existing constitutive models, on the one hand conventional SMA models and on the other hand also extended models considering fatigue behaviour.
- Comparison of two or three existing models capturing the described effects and highlighting their advantages and disadvantages.
- Discussion of engineering implications, including strategies by using the material models for designing SMA components subjected to cyclic actuation.

Expected Outcome: By completing this task, students will gain:

- A deeper understanding of the thermomechanical behaviour of SMAs.
- Insight into fatigue mechanisms and their impact on functional performance.
- The ability to critically assess the limitations of current models and propose directions for improvement.

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